Comenius University in Bratislava
Faculty of Mathematics, Physics and Informatics

Exploiting artificial intelligence in software documentation and comprehension

Master's Thesis

2026
Bc. Adelina Ivantaeva

Comenius University in Bratislava
Faculty of Mathematics, Physics and Informatics

Exploiting artificial intelligence in software documentation and comprehension

Master's Thesis

Study Programme: Applied Computer Science
Field of Study: Computer Science
Department: Department of Applied Informatics
Supervisor: Ing. Lukáš Radoský

Bratislava, 2026
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ZADANIE ZÁVEREČNEJ PRÁCE

Meno a priezvisko študenta:	Bc. Adelina Ivantaeva
Študijný program:	aplikovaná informatika (Jednoodborové štúdium, magisterský II. st., denná forma)
Študijný odbor:	informatika
Typ záverečnej práce:	diplomová
Jazyk záverečnej práce:	anglický
Sekundárny jazyk:	slovenský

Názov: Exploiting artificial intelligence in software documentation and comprehension
Využitie umelej inteligencie v dokumentácii a porozumení softvéru

Anotácia: Nástup jazykových modelov vyvolal veľký záujem o ich využitie vo vývoji softvéru. Veľkú pozornosť prilákala najmä ich schopnosť generovať zdrojový kód. Po úvodnom nadšení sa pozornosť presúva viac k ich využitiu v iných fázach cyklu vývoja softvéru (SDLC), ako je analýza, návrh, testovanie, či údržba.

Zvoľte si jednu či viacero úloh orientovaných na dokumentáciu alebo pochopenie softvéru. Preskúmajte možnosti využitia umelej inteligencie pri týchto úlohách, najmä však využitie jazykových modelov. Implementujte riešenie zvolenej podúlohy či podúloh využívajúce umelú inteligenciu. Svoje riešene vyhodnoťte na vhodne zvolenej vzorke dát alebo používateľov. Porovnajte s existujúcimi riešeniami.

Cieľ: Návrh a implementácia riešení dokumentácie a pochopenia softvéru pomocou umelej inteligencie.

Literatúra: [1] Xinyi Hou, Yanjie Zhao, Yue Liu, Zhou Yang, Kailong Wang, Li Li, Xiapu Luo, David Lo, John Grundy, and Haoyu Wang. 2024. Large Language Models for Software Engineering: A Systematic Literature Review. ACM Trans. Softw. Eng. Methodol. 33, 8, Article 220 (November 2024), 79 pages. https://doi.org/10.1145/3695988
[2] Christof Tinnes, Alisa Welter, and Sven Apel. 2025. Software Model Evolution with Large Language Models: Experiments on Simulated, Public, and Industrial Datasets. Proceedings of the IEEE/ACM 47th International Conference on Software Engineering. IEEE Press, 950–962. https://doi.org/10.1109/ICSE55347.2025.00112
[3] Jules White, Quchen Fu, Sam Hays, Michael Sandborn, Carlos Olea, Henry Gilbert, Ashraf Elnashar, Jesse Spencer-Smith, and Douglas C. Schmidt. 2023. A Prompt Pattern Catalog to Enhance Prompt Engineering with ChatGPT. In Proceedings of the 30th Conference on Pattern Languages of Programs (PLoP '23). The Hillside Group, USA, Article 5, 1–31.

Kľúčové slová: umelá inteligencia, jazykové modely, softvérová dokumentácia, pochopenie softvéru , cyklus vývoja softvéru

Vedúci:	Ing. Lukáš Radoský
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Spôsob sprístupnenia elektronickej verzie práce:
bez obmedzenia

Dátum zadania: 26.11.2025

Dátum schválenia: 26.11.2025

prof. RNDr. Roman Ďurikovič, PhD.
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THESIS ASSIGNMENT

Name and Surname:	Bc. Adelina Ivantaeva
Study programme:	Applied Computer Science (Single degree study, master II. deg., full time form)
Field of Study:	Computer Science
Type of Thesis:	Diploma Thesis
Language of Thesis:	English
Secondary language:	Slovak

Title: Exploiting artificial intelligence in software documentation and comprehension

Annotation: The advent of language models sparked great interest in their use in software development. Their ability to generate source code attracted particular attention. After the initial enthusiasm, attention is shifting more towards their use in other phases of the software development life cycle (SDLC), such as analysis, design, testing, and maintenance.

Choose one or more tasks oriented towards software documentation or comprehension. Explore the possibilities of using artificial intelligence in these tasks, especially the use of language models. Implement a solution for the selected subtask or subtasks based on artificial intelligence. Evaluate your solution on a suitable sample of data or users. Compare with existing solutions.

Aim: Design and implementation of software documentation and comprehension solutions based on artificial intelligence.

Literature:
[1] Xinyi Hou, Yanjie Zhao, Yue Liu, Zhou Yang, Kailong Wang, Li Li, Xiapu Luo, David Lo, John Grundy, and Haoyu Wang. 2024. Large Language Models for Software Engineering: A Systematic Literature Review. ACM Trans. Softw. Eng. Methodol. 33, 8, Article 220 (November 2024), 79 pages. https://doi.org/10.1145/3695988
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[3] Jules White, Quchen Fu, Sam Hays, Michael Sandborn, Carlos Olea, Henry Gilbert, Ashraf Elnashar, Jesse Spencer-Smith, and Douglas C. Schmidt. 2023. A Prompt Pattern Catalog to Enhance Prompt Engineering with ChatGPT. In Proceedings of the 30th Conference on Pattern Languages of Programs (PLoP '23). The Hillside Group, USA, Article 5, 1–31.

Keywords: artificial intelligence, large language models, software documentation, software comprehension, software development life cycle, SDLC

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Assigned: 26.11.2025

Approved: 26.11.2025 prof. RNDr. Roman Ďurikovič, PhD.
Guarantor of Study Programme

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Student Supervisor

Acknowledgments: Here you can thank your supervisor and/or other persons who somehow helped you with this thesis. You should acknowledge grants/projects in case your research was financed by any.

Abstract

English abstract: 100-500 words; keep it formatted in one paragraph. Do not use abbreviations and other niche advanced terms here, which will be explained in the thesis later.

Keywords: first, second, third, (fourth, fifth)

Abstrakt

The Slovak abstract: should be the translation of the English one.

Kľúčové slová: jedno, druhé, tretie (prípadne štvrté, piate)

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Choi, Y., Na, C., Kim, H., and Lee, J.-H. (2023). Readsum: Retrieval-augmented adaptive transformer for source code summarization. *IEEE Access*, 11:51155–51165.

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Nam, D., Macvean, A., Hellendoorn, V., Vasilescu, B., and Myers, B. (2024). Using an llm to help with code understanding. In *2024 IEEE/ACM 46th International Conference on Software Engineering (ICSE)*, pages 1184–1196.

Veerappa Dinesh, S. E., M, S. T., K, S., Nayagam, M. G., and Amanullah, M. (2025). Llm-augmented software documentation generator with contextual memory. In *2025 IEEE 6th Global Conference for Advancement in Technology (GCAT)*, pages 1–6.

Yang, D., Simoulin, A., Qian, X., Liu, X., Cao, Y., Teng, Z., and Yang, G. (2025). DocAgent: A multi-agent system for automated code documentation generation. In Mishra, P., Muresan, S., and Yu, T., editors, *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, pages 460–471, Vienna, Austria. Association for Computational Linguistics.